

A candidate classification of prime triangle-tiling counts

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Abstract

We present a candidate proof that a prime p is the number of congruent triangles in a dissection of some triangle if and only if $p = 2$, $p = 3$, or $p \equiv 1 \pmod{4}$. The two new proposed geometric inputs exclude the primitive scale-one targets in the scalene W and β -isosceles branches for tiles satisfying $3\alpha + 2\beta = \pi$. The isosceles argument forces a standard corner whose complement is the forbidden W triangle; the W argument uses a column induction. Both permit arbitrary T-junctions. We give the complete passage from these inputs and the published shape classification to the prime statement, including explicit existence constructions and the arithmetic for the other branches. The classification of all composite counts remains unresolved.

Status and contribution. This is a candidate proof, released for mathematical scrutiny. Denis Paliy set the research direction and guided the investigation; ChatGPT assisted with exploring proof strategies, drafting arguments, and carrying out internal checks. The draft has not received external peer review or formal verification. No research-priority claim is made. The forced-patch and column-completion arguments are the principal new steps requiring review.

1 The prime case and its dependencies

A tiling is a finite dissection into triangles with pairwise disjoint interiors, all congruent to one fixed tile. Reflections are allowed, and an edge may meet vertices of several other tiles: no edge-to-edge assumption is imposed.

Proposed prime-classification theorem. For a prime p , there exists a triangle tiled by p congruent triangles if and only if

$$p = 2, \quad p = 3, \quad \text{or} \quad p \equiv 1 \pmod{4}.$$

The proposed new ingredients are exactly the two scale-one exclusions proved in Section 2 and Appendix A. This section explains all outside dependencies and the full deduction from those ingredients. It does not infer a classification of composite counts from the prime case.

1.1 Published inputs and exhaustive branch list

The shape classification is due to Laczkovich [1], Theorems 4.1, 5.1 and 5.3. Its isosceles specialization is recorded in [3], Theorem 3.1 and Table 1; its non-isosceles specialization is restated in [7], Theorems 6 and 9, with an independent comparison in [6], Table 1. The reptiling classification is due to Snover, Waiveris and Williams [2]; see also [3], Theorem 2.1.

We use the specified versions of the sources. In particular, [4] means version 4: its corrected necessary equations are used without assuming that a divisibility condition sufficient for a

particular construction is necessary for every tiling. Rationality in the irrational 120° branches is supplied by Beeson–Zhang [6], Theorem 1.1, rather than the flawed older rationality argument.

Suppose $p > 3$ is prime and a tiling exists. If the tile is similar to the target, the reptiling classification gives a square, three times a square, or a sum of two integer squares as the count. Only the last option can equal $p > 3$, and reduction modulo 4 then gives $p \equiv 1 \pmod{4}$. For nonsimilar tilings the following list is exhaustive. Here “Group 1” denotes $3\alpha + 2\beta = \pi$.

Target and tile branch	Exclusion for $p > 3$
Equilateral; rational tile angles	Elementary area calculation below.
Equilateral; irrational angles with a 60° tile angle	[5], Theorem 3(i), and the factorization below.
Equilateral; irrational angles with a 120° tile angle	[5], Theorem 6, with rationality from [6].
Non-equilateral isosceles; right tile	[3], Theorem 7.8: the count is square or even.
Non-equilateral isosceles; tile $\gamma = 2\alpha$	[3], Theorem 11.7: the count is not squarefree.
Group 1; target $(2\alpha, \alpha, 2\beta)$	[4], Theorem 16.
Group 1; isosceles base angle $\alpha + \beta$	[4], Theorem 22.
Group 1; isosceles base angle α	[4], Theorem 26.
Group 1; $W = (2\alpha, \beta, \alpha + \beta)$	Integer-scale reduction below, then Appendix A.
Group 1; isosceles base angle β	Integer-scale reduction below, then Section 2.
Irrational 120° tile; non-equilateral isosceles target	Signed-direction obstruction below.
Irrational 120° tile; non-isosceles target	The four sine-law factorizations below.

To see exhaustiveness, first remove reptilings. The equilateral classification gives precisely the five tile types covered by the first three rows. For a non-equilateral isosceles target, the classification gives right tiles, the $\gamma = 2\alpha$ family, the three isosceles Group 1 shapes, and $\gamma = 120^\circ$. For a non-isosceles target with rational multiples of π as angles, the rational-angle theorem permits only a reptiling. In the remaining case the tile also has an irrational angle, because target angles are sums of tile angles; the classification gives the two scalene Group 1 shapes and the four scalene 120° shapes. Labels may be permuted as allowed by these classifications; the relation $3\alpha + 2\beta = \pi$ is not silently swapped.

Source qualifications. No broad prime-count conclusion from [7] is used: its original Group 1 and isosceles 120° dependencies are insufficient. In particular, we do not use its Theorem 3, Theorem 5(iii), Theorem 22 or Corollary 23 as an established global result. The withdrawn paper arXiv:2607.19572 is not used. Two smaller wording errors also require care: Corollary 7.9 of [3] omits the valid $N = 2$ exception, so we use Theorem 7.8 for odd primes; and the rational-angle equilateral discussion omits the $2k^2$ family of 30–60–90 tiles, which is included below.

1.2 Direct integer-scale reduction in the two new branches

Rationality and primitive normalization in Group 1 give

$$(a, b, c) = (uv, v^2 - u^2, v^2), \quad 0 < u < v, \quad \gcd(u, v) = 1.$$

Write $Q = 2v^2 - u^2$ and $P = 3v^2 - u^2$. The sine law gives the side proportions

$$W : (v^3, uQ, v(v^2 - u^2)), \quad \beta\text{-isosceles} : (v^3, v^3, uP).$$

Both triples are primitive: $\gcd(v^3, uQ) = \gcd(v^3, uP) = 1$ follows from $\gcd(u, v) = 1$. Every actual target side is an integer sum of integer tile-edge lengths. Bézout's identity therefore makes its similarity scale a positive integer j . Comparison of areas yields

$$N = j^2 Q \quad \text{for } W, \quad N = j^2 P \quad \text{for the } \beta\text{-isosceles shape.}$$

This direct argument supplies the required integer scale independently of coloring-divisibility claims. It agrees with the corrected formulas in [4], Corollary 1 and Theorem 19. Each of P, Q exceeds one, so a prime count forces $j = 1$. The proposed geometric theorems below exclude exactly these two targets at $j = 1$, for every admissible u, v . Neither uniqueness of representation by these quadratic forms nor a construction-specific condition such as $v \mid u^2$ is needed.

1.3 Classical equilateral cases and the irrational 60° case

For equilateral tiles the count is a square. For tile sides $(1, 1, \sqrt{3})$, an equilateral target side has the form $L = A + B\sqrt{3}$ with nonnegative integers A, B ; its area ratio is $N = L^2$. Rationality forces $AB = 0$, giving a square or three times a square. For tile sides $(1, \sqrt{3}, 2)$, the ratio is $N = L^2/2$; again $AB = 0$, and integrality gives $N = 2k^2$ or $6k^2$. None of these counts is prime greater than three.

In the irrational 60° branch, [5], Theorem 3(i), gives integers M, y satisfying

$$y^2 = (9p - M^2)(p - M^2), \quad 0 < M^2 < p.$$

Set $X = 5p - M^2$. Then $(X - y)(X + y) = 16p^2$, with both factors positive. They cannot both be divisible by p , because their sum is $10p - 2M^2$ and $p \nmid M$. Thus one contains the full factor p^2 , so their sum is greater than p^2 . But the sum is less than $10p$, which is impossible for $p \geq 11$. For $p = 5$ the candidate discriminants are 176, 41; for $p = 7$ they are 372, 177. None is a square. This checks every prime $p > 3$ without relying on the printed calculation in part (ii) of that theorem.

1.4 The irrational 120° branches

By rationality, normalize the tile to primitive positive integers a, b, c with

$$c^2 = a^2 + ab + b^2.$$

The three integers are pairwise coprime. Also $3 \nmid c$: if $3 \mid c$, reduction modulo 3 gives $a \equiv b \pmod{3}$; writing $a = b + 3h$ yields $c^2 = 3b^2 + 9bh + 9h^2$, which forces $3 \mid b$ and contradicts primitiveness.

For the non-isosceles targets, the sine law gives the following primitive integer side proportions and area ratios. Thus the actual similarity scale λ is a positive integer by the same boundary length and Bézout argument as above.

Target angles	Primitive side proportions	N
$(\alpha, 2\alpha, 3\beta)$	$(c^2, c(a+2b), 3b(a+b))$	$3\lambda^2(a+2b)(a+b)$
$(\alpha, 2\beta, 2\alpha+\beta)$	$(ac, b(2a+b), (a+b)c)$	$\lambda^2(2a+b)(a+b)$
$(\alpha, \alpha+\beta, \alpha+2\beta)$	$(a, c, a+b)$	$\lambda^2(a+b)/b$
$(2\alpha, 2\beta, \alpha+\beta)$	$(a(a+2b), b(2a+b), c^2)$	$\lambda^2(a+2b)(2a+b)$

The first, second and fourth expressions are composite. In the third, $b \mid \lambda^2$, and $a+b$ is composite: if $a+b=q$ were prime, the cosine equation modulo q would give $q \mid (c-a)(c+a)$. Since $a < c < a+b=q$, the first factor cannot contain q ; hence $c+a=q$, giving $c=b$, a contradiction. These are the individual computations of [7], Theorems 18–21, without that paper’s global conclusion.

For a non-equilateral isosceles target, take the base angle to be α after interchanging a, b if necessary. Its primitive side proportions are $(c, c, a+2b)$: any prime common to c and $a+2b$ would divide $3b^2$, and $3 \nmid c$. Thus its sides are $k(c, c, a+2b)$ for an integer $k > 0$, and

$$N = \frac{k^2(a+2b)}{b}.$$

Because $\gcd(b, a+2b) = 1$, integrality and primality force $k^2 = b$ and $N = a+2b$.

We reproduce the signed-direction obstruction associated with Bonfioli [8]. Fix a reference edge. Across a shared edge segment, directions propagate through angle differences of the tile; the adjacency graph of a finite dissection is connected. All directed edge directions therefore lie, after an overall rotation, in $G = \mathbb{Z}\pi/3 + \mathbb{Z}\alpha$ modulo 2π . Irrationality of α/π makes the character

$$f(j\pi/3 + \ell\alpha) = (-1)^j$$

well defined. Give an oriented segment of length L the weight $Lf(\text{direction})$. Reversal negates its weight. Subdivision at every intersection and T-junction cancels all interior subsegments in the sum of oriented tile boundaries. Each tile contributes $\pm(c+a-b)$, while the target contributes $k(a+2b-2c)$. Consequently

$$\frac{k(a+2b-2c)}{c+a-b} = \frac{c-a-b}{k} \in \mathbb{Z},$$

where the identity uses $k^2 = b$ and the cosine equation. Hence $k \mid (a+b-c)$. But $a < c < a+k^2$, so $c = a+kr$ for an integer $1 \leq r \leq k-1$. Substitution gives

$$a(2r-k) = k(k-r)(k+r).$$

Since $\gcd(a, k) = 1$, we have $k \mid 2r$. The only possible multiple in $2 \leq 2r \leq 2k-2$ is $2r = k$, making the left side zero and the right side positive. For $k = 1$ there is no possible r . This excludes the isosceles branch, including T-junction tilings.

The equilateral irrational 120° branch is excluded by [5], Theorem 6, with the rationality input stated above. This completes all 120° rows.

1.5 Existence and completion of the conditional deduction

For $p = 2$, cut an isosceles triangle along its altitude. For $p = 3$, join the center of an equilateral triangle to its vertices; the pieces are three congruent 120° isosceles triangles.

For $p \equiv 1 \pmod{4}$, the sum-of-two-squares theorem gives $p = m^2 + n^2$ with positive integers m, n . Let the tile T be the right triangle with legs m, n . Take a larger right triangle OAB with

$$OA = m\sqrt{p}, \quad OB = n\sqrt{p}, \quad AB = p.$$

If H is the foot of the altitude from O to AB , the projection identities give

$$AH = m^2, \quad BH = n^2, \quad OH = mn.$$

Thus OAH and OBH are copies of T enlarged by factors m and n . Their standard parallel-line subdivisions contain m^2 and n^2 triangles congruent to T , filling OAB with exactly p tiles.

Every nonsimilar branch has now either been excluded by a stated published input and the calculations above, or reduced to one of the two proposed geometric exclusions below. With those exclusions, a prime $p > 3$ allowing a tiling must be a reptiling count and satisfy $p \equiv 1 \pmod{4}$. The constructions prove the converse. The remaining mathematical burden of this draft is therefore concentrated in the two scale-one proofs that follow.

2 The isosceles obstruction

2.1 Main theorem and its geometric idea

Let $0 < u < v$, $\gcd(u, v) = 1$, and put

$$a = uv, \quad b = v^2 - u^2, \quad c = v^2, \quad P = 3v^2 - u^2, \quad Q = 2v^2 - u^2.$$

Let the congruent tile have sides (a, b, c) opposite angles (α, β, γ) . Then $3\alpha + 2\beta = \pi$ and $\gamma = 2\alpha + \beta > \pi/2$.

Theorem. The isosceles triangle with sides (v^3, v^3, uP) cannot be tiled by these tiles. Its area would require exactly P tiles. The triangle with sides (v^3, uQ, vb) cannot be tiled by these tiles either; its area would require Q tiles.

The second assertion is the scale-one W theorem, proved in the appendix. The argument below reduces the first assertion to the second by an actual dissection: starting at the three- α apex, it forces a standard v^2 -tile corner. Its complement is exactly the forbidden W triangle.

These are statements about the full scale-one families. They do not classify larger-scale composite tilings or by themselves solve every part of Erdős problem 634.

2.2 Elementary angular and arithmetic facts

The number α/π is irrational: $2 \cos \alpha = 2 - u^2/v^2$ is rational strictly between 1 and 2, whereas for a rational multiple of π it would be a rational algebraic integer, hence an integer. Consequently the tile-angle inventories at a β corner, an α sector, a $\theta = \alpha + \beta$ sector, and a 3α apex are respectively one β , one α , one α plus one β , and three α sectors.

We need three exact positive-chain facts. All coefficients count whole edges and are nonnegative integers.

$$ib = Aa + Bb + Cc, \quad 0 < i < v \implies A = C = 0, \quad B = i. \quad (1)$$

Indeed $B \equiv i \pmod{v}$ and $0 \leq B \leq i$.

$$ia = Aa + Bb + Cc, \quad 0 < i < v \implies B = C = 0, \quad A = i. \quad (2)$$

Write $B = v\ell$ and $\ell v + C = Tu$; division of the length equation gives $i = A + (T - \ell)v + \ell(v - u)$. If $\ell + C > 0$, then $Tu = \ell v + C > \ell u$, so $T > \ell$ and $i \geq v$, impossible.

$$\begin{aligned} nc &= Aa + Bb + Cc, \quad 0 < n < v \\ \implies B &= 0, \quad A = vh, \quad C = n - uh \quad \text{for an integer } h \geq 0. \end{aligned} \tag{3}$$

To prove (3), write $B = v\ell$ and $d = v - u$. The equation gives

$$A = vh - d\ell, \quad n = C + uh + d\ell.$$

If $\ell > 0$, nonnegativity of A forces $h \geq 1$, and then $n \geq u + d = v$. Thus $\ell = 0$. In particular, when $n < u$, (3) also forces $A = 0$ and $C = n$.

The distinction is crucial: below height u a pure- c seam has only c opposite it; below height v it may have a opposite it, but it has **no b opposite it**. The downward apex induction needs the latter, stronger range.

Throughout the proof the tiling is fixed. A collinear component means a connected union of actual whole tile edges on one supporting line. An internal maximal component has a complete chain on each side: an edge cannot end inside an edge on the other side without another collinear edge continuing the remaining straight sector. Nor can a supplied next edge be swallowed by a tile at a junction. Maximality prevents an endpoint from lying inside any constituent edge. When the proof instead uses a subsegment, it separately establishes endpoints for both chains, using external supporting lines or the starting edge of an actual tile. Thus none of the length lemmas is applied to a chain of merely partial edges.

2.3 Equal-side boundary fact and coordinates

No corner of the isosceles target contains a γ sector: its two base corners contain one β each, and its apex contains exactly three α sectors. This remains true when the total apex angle 3α exceeds γ .

Every a - or b -edge on a target side contributes a γ endpoint, and two such endpoints cannot coincide at a straight boundary junction because $2\gamma > \pi$. A side consisting of n edges has only $n - 1$ internal junctions. Hence every side contains at least one c -edge.

An equal side has length $v^3 = vc$. Reduction modulo v makes its b -count a multiple of v . If it had at least v b -edges, removing v of them would leave a positive whole-edge decomposition of

$$vc - vb = ua.$$

Since $u < v$, (2) makes every remaining edge an a -edge, contradicting the presence of a c -edge. Therefore **both equal sides have no b -edge**.

Place the target at $A = 0$, $B = vq$, $C = uP$, where $p = a$ is horizontal, $q = ce^{i\beta}$, and $q - p = be^{i\theta}$. Put $Z(i, t) = ip + tq$. Use

$$\begin{aligned} D(i, t) &= \text{conv}(Z(i, t), Z(i + 1, t), Z(i, t + 1)), \\ U(i, t) &= \text{conv}(Z(i + 1, t), Z(i, t + 1), Z(i + 1, t + 1)). \end{aligned}$$

Their angles at the listed vertices are respectively (β, γ, α) and (α, γ, β) .

The apex has exactly three α tiles. Its two outer edges are c , because the equal sides have no b . Each α tile has incident sides b, c , so its two inner radial seams comprise one b/b seam and one b/c seam. Choose the reflected labeling so the latter is next to the left outer tile $D(0, v - 1)$.

2.4 Reverse-apex state

Write $V_m = Z(m, v - m)$ for $0 \leq m \leq v$. Let $T(m, t)$ be the actual standard truncated corner region consisting of:

- all $D(i, s)$ with $0 \leq i < m$, $s \geq t$, and $i + s \leq v - 1$;
- all $U(i, s)$ with $0 \leq i < m$, $s \geq t$, and $i + s \leq v - 2$.

Geometrically this is the standard subdivision of the polygon with oblique coordinates

$$(0, t), \quad (m, t), \quad (m, v - m), \quad (0, v),$$

when $t \leq v - m$, with zero-height edges allowed.

For each $m = 1, \dots, v - 1$, the apex diagonal b -chain $B = V_0, V_1, \dots, V_m$ will be actual. Its opposite chain starts at B with the central apex c -edge. Since $m < v$, short b purity implies that V_m is in the relative interior of an actual opposite edge. Thus the positive q -ray from V_m enters that opposite tile's interior.

Let M_m be the actual maximal collinear component from V_m in direction $-q$. Its upper end is V_m , by the preceding crossing. We prove:

1. Its whole left side is a chain of normal c -edges of U tiles.
2. If its lower endpoint is $Z(m, r)$, then $T(m, r)$ is actual; more generally every reached integer height t supplies $T(m, t)$.
3. Its opposite side has no b -edge.

These statements are proved in increasing m ; within each column, the continuation goes downward. Earlier columns may terminate earlier, but whenever a newly supplied U extends one of them downward, its already proved conditional completion statement applies. No assumption is made that any column reaches the base.

2.5 Reverse horizontal cap

The first outer apex tile is $D(0, v - 1)$. Its b -edge is BV_1 , and the opposite central c -edge starts at B and overruns V_1 since $c > b$.

More generally, suppose the diagonal has just reached V_m , where $m < v$, and $T(m - 1, v - m)$ is known. For $m = 1$, the old region $T(0, v - 1)$ is empty and the new tile $D(0, v - 1)$ is the outer apex tile. The new $D(m - 1, v - m)$ gives, together with the old region, a horizontal chain of m a -edges from $(v - m)q$ to V_m .

Its negative extension at the outer side leaves the convex target. Its positive extension at V_m enters the crossing opposite tile. Thus the opposite chain is a complete chain of total length ma , hence exactly m a -edges.

At V_m the already placed D contributes γ and the crossing tile a straight sector π , so the remaining sector is $\theta = \alpha + \beta < \gamma$. Consequently the opposite a -tile has β , not γ , at V_m ; it is $U(m - 1, v - m - 1)$. Propagating left, every successive opposite a -tile must also be a normal U : a reversed one would put two γ sectors in the same lower half-plane at a shared junction.

At the outer boundary point $(v - m)q$, the old D contributes β and the new U contributes γ , leaving α . The next boundary edge is c , since that equal side has no b ; this gives $D(0, v - m - 1)$.

At each interior grid point $Z(i, v - m)$, where $1 \leq i < m$, the existing upper row and the two new U tiles leave exactly one α sector. The new left U supplies an actual downward c -edge in the earlier column M_i , connected to its upper component through the old actual region. Its

already proved no-opposite- b property excludes b on the right of that q -edge. Hence its α tile uses c and is $D(i, v - m - 1)$.

Explicitly, at a reached height t , the old region contains the collinear c -edges of $U(i - 1, s)$ for $t \leq s \leq v - i - 1$, joining $Z(i, t)$ to V_i . The new edge of $U(i - 1, t - 1)$ attaches to this very component. If it extended below an earlier claimed endpoint, that would contradict the already established maximality; it would not create a different column. The same connectivity applies in the downward continuation below.

This supplies $T(m, v - m - 1)$ and the first complete c -edge of M_m .

2.6 Downward column completion

Assume $T(m, t)$ is actual, with $t \leq v - m - 1$, and the real component M_m continues downward from $X = Z(m, t)$. Its old left-side normal tiles $D(m - 1, t)$ and $U(m - 1, t)$ contribute $\gamma + \alpha = \pi - \beta$. The actual downward q -ray therefore leaves one β tile on its left.

That β tile has either a or c on the downward q -ray. If it had a , its other incident side would be a c -edge from X horizontally to the left. But the actual upper horizontal chain from tq to X consists of m a -edges. Starting at X with this c -edge, the opposite chain must cover the whole upper chain: at a T-junction inside an upper edge a straight sector forces continuation, and an upper-chain grid endpoint cannot let the actual next upper edge be swallowed by a tile. At tq the external side prevents overrun. Hence the lower chain would be a complete positive chain of length ma containing a c -edge, contrary to short a purity, since $m < v$.

Thus the β tile has c on the q -ray and is $U(m - 1, t - 1)$. Its top a -edge starts a whole lower chain along the existing upper chain of length ma . Short a purity makes the lower chain exactly m a -edges. Starting at X , the same two- γ argument forces all these lower tiles to be normal $U(i, t - 1)$.

At tq , the old $D(0, t)$ contributes β and $U(0, t - 1)$ contributes γ , leaving α , so absence of boundary b gives $D(0, t - 1)$. At each inner point $Z(i, t)$, where $1 \leq i < m$, exactly one α remains. The new actual downward c -edge is on the earlier column M_i ; its no-opposite- b property makes that α tile $D(i, t - 1)$. Therefore $T(m, t - 1)$ is actual.

Every actual continuation adds one complete normal c -edge. Since the target is bounded, the component ends at some grid height $r \geq 0$. Its length is nc , where

$$n = (v - m) - r, \quad 0 < n \leq v - m < v.$$

Arithmetic fact (3) gives no b -edge on the opposite side. This closes the induction for M_m .

2.7 Advancing the apex diagonal

At V_m the crossing opposite tile, the old $D(m - 1, v - m)$, and the first $U(m - 1, v - m - 1)$ leave one α sector immediately on the right of the downward q -ray. Its q -side cannot be b by M_m 's just established no-opposite- b property. It is therefore c , and the tile is $D(m, v - m - 1)$. Its b -edge extends the actual diagonal to V_{m+1} .

For $m + 1 < v$, short b purity again says that this endpoint lies inside an actual opposite edge. The reverse horizontal cap then initializes the next column. For $m = v - 1$ the same forcing gives $D(v - 1, 0)$ and reaches $V_v = vp$ on the base; no short b statement with index v is used.

The column M_{v-1} starts at height 1 and its initialization reaches height 0. Its completion statement gives $T(v - 1, 0)$. Adding $D(v - 1, 0)$ gives the entire standard

$$K_v = \text{conv}(0, vp, vq),$$

with its v^2 tiles.

In particular the base begins at A with v consecutive a -edges, of total length $L = va = uc$.

2.8 The complement is a forbidden W triangle

Let $L = vp$. The forced K_v has vertices A, B, L , consists of v^2 whole tiles, and is a genuine corner of the target. Its diagonal BL is the actual chain of v b -edges. Thus removing its tiles leaves a tiling of the triangle BLC .

But

$$BC = v^3, \quad BL = vb, \quad LC = uP - va = u(2v^2 - u^2) = uQ.$$

The remaining tile count is $P - v^2 = Q$. Hence the complement is exactly the scale-one W triangle excluded below. This proves the isosceles assertion. No base-edge count, two- c boundary lemma or assumption about an ordering of the base edges is needed. $\square$

A Complete scale-one W exclusion

The following proof is reproduced to make the result self-contained. It uses a different orientation of coordinates, reset in its statement. The arithmetic facts called S_b, S_a, S_c below are respectively (1), (2), and the $n < u$ consequence of (3).

A.1 Statement and coordinate convention

Let $0 < u < v$, $\gcd(u, v) = 1$, and put

$$a = uv, \quad b = v^2 - u^2, \quad c = v^2, \quad Q = 2v^2 - u^2.$$

The theorem is that the triangle with side lengths

$$(v^3, uQ, vb)$$

cannot be tiled by Q congruent copies of (a, b, c) . Its angles are $(\beta, 2\alpha, \alpha + \beta)$, where $3\alpha + 2\beta = \pi$.

Name the β corner O . Direct OA horizontally to the right, with $|OA| = v^3$. The other side OC has length $u(b + c) = uQ$ and makes angle β with OA . Put

$$p = a, \quad q = ce^{i\beta}, \quad O = 0, \quad A = v^3, \quad C = \frac{u(b + c)}{c}q, \quad Z_{i,t} = ip + tq.$$

Thus $q - p = be^{i\theta}$, where $\theta = \alpha + \beta$. The q direction is along the side OC , whose edge counts will be $(0, u, u)$. It is **not** assumed to be a pure c -grid initially.

The two adjacent side lengths satisfy the strict inequalities

$$\begin{aligned} |OA| &= v^3 > ua = u^2v, \\ |OC| &= u(b + c) > uc. \end{aligned} \tag{L}$$

Consequently every corner front of index $j \leq u$ has both endpoints strictly inside the two external sides, so its opposite side is an internal chord. This excludes the degenerate situation in which a purported front is already the outer third side of the target.

Use the standard actual triangles

$$D_{i,t} = \text{conv}(Z_{i,t}, Z_{i+1,t}, Z_{i,t+1}), \quad U_{i,t} = \text{conv}(Z_{i+1,t}, Z_{i,t+1}, Z_{i+1,t+1}).$$

The angles of D at its listed vertices are (β, γ, α) ; those of U are (α, γ, β) . Let $R(m, n)$ denote the actual rectangle of all such tiles for $0 \leq i < m$, $0 \leq t < n$.

A.2 Boundary counts, corner, and short-chain arithmetic

Every target angle is less than γ . Thus no boundary corner can contain a γ sector. Each boundary a - or b -edge contributes a γ endpoint. Two γ sectors cannot share a straight boundary junction. Consequently every target side has at least one c -edge.

On OC , write its counts as (A_0, B_0, C_0) , with $C_0 \geq 1$. The length equation is

$$A_0uv + B_0(v^2 - u^2) + C_0v^2 = u(2v^2 - u^2).$$

Reduction modulo v gives $B_0 = u + vk$, $k \geq 0$. Write $C_0 = u\ell - vk$; then $A_0 = uk + v(1 - \ell)$. For $k \geq 1$, $C_0 \geq 1$ requires $\ell \geq k + 1$, but $A_0 \geq 0$ requires $\ell \leq k$. Hence $k = 0$, and positivity gives $\ell = 1$. Therefore

$$\boxed{(A_0, B_0, C_0) = (0, u, u).} \quad (\text{B})$$

At O , the unique β tile has incident sides a, c . Since OC has no a -edge, its first edge is c and OA starts with a . Thus $D_{0,0}$ is present.

We use S_b , S_a and S_c for the positive-chain facts (1), (2), and the $n < u$ consequence of (3), proved above. The exact angle inventories and irrationality of α/π were also established above.

A.3 First column: actual c-only strip, then no right a-edge

The internal b -edge of $D_{0,0}$ is the chord joining p and q . Both endpoints are on the outer boundary. Its opposite chain consists of a single b -edge by S_b . The opposite tile is $U_{0,0}$: the other orientation would put two γ sectors at the boundary point p .

Let L_1 be the actual maximal connected collinear component of tile boundaries from p in direction q . Its negative extension leaves the target across OA . Its left side starts with the c -edge of $U_{0,0}$.

The distance from this line to the supporting line OC is $a \sin \beta = h_c$, the tile altitude to c . A tile attached on the left with an a - or b -edge would have larger altitude and cross the external supporting line. Hence **every complete edge on the entire left of L_1 has length c** , even at arbitrary T-junctions.

The component is finite. Maximality means its left edges form a complete chain starting at p and ending at $p + nq$, for some integer $n \geq 1$. There is no possibility of ending inside one of these edges, because that edge would extend the component.

For each left c -edge from $p + iq$ to $p + (i + 1)q$, its opposite vertex lies on the supporting line OC , since its altitude equals the strip width. The two possible vertex positions are

$$(i + 1)q, \quad (i + \lambda)q, \quad \lambda = 2\frac{a}{c} \cos \beta = \frac{u^2(3v^2 - u^2)}{v^4}.$$

The second point has signed distance

$$(i + \lambda)c = iv^2 + \frac{u^2(3v^2 - u^2)}{v^2}$$

from O , which is nonintegral because $\gcd(u, v) = 1$. An actual tile vertex on OC must be a boundary junction, and every boundary-junction distance is an integer sum of tile-edge lengths. Thus the second point is impossible. All left tiles are the normal $U_{0,i}$.

The spaces between consecutive U tiles and OC are the triangles $D_{0,i}$. They are bounded by actual tile edges and the external boundary and each has exactly one tile's area. They must therefore be single tiles, so $R(1, n)$ is present. Equivalently, this can be checked by successive boundary angle fans. The original $D_{0,0}$ is already present.

At nq the tiles $D_{0,n-1}$ and $U_{0,n-1}$ contribute α and γ , leaving β in the target boundary half-plane. This point cannot be C : its two existing angles already total $\alpha + \gamma > \alpha + \beta$, the target angle at C . It cannot lie beyond C either. Hence it is an internal boundary junction, and its next

boundary edge belongs to a β tile. That edge is a or c ; by (B), it is c . Therefore the existing first n boundary c -edges require one additional c -edge, giving

$$\boxed{n < u.} \tag{H}$$

The opposite side of L_1 is a complete edge chain of length nc . By S_c , every edge on it is also c . In particular, there is **no right a -edge on L_1** .

The same argument gives the first-column rectangular completion property: whenever this actual component reaches height t , $R(1, t)$ is present. If $u = 1$, (H) contradicts $n \geq 1$ already; hence that case is excluded at this point.

A.4 Horizontal cap without a pre-assumed c-grid

Suppose $R(i, t)$ and $D_{i,t-1}$ are present, $0 < i < v$, and an actual θ -directed edge contains $X = Z_{i,t}$ in its relative interior, with its tile on the lower/right side.

The rectangle supplies a horizontal chain of i lower a -edges from tq to X . The negative extension at tq leaves the target across OC , and the positive extension at X enters the crossing tile's interior. Hence this is the entire actual horizontal component. Its complete upper chain has length ia , so S_a makes it exactly i a -edges.

At tq the existing α and γ sectors leave β . The point is an internal point of OC for the same reason as in (H). Since OC has no a -edges, its next actual edge must be c , giving $D_{0,t}$. At every later upper-chain junction, the preceding normal D contributes γ . The remaining upper angle is θ , and $\gamma > \theta$. The next upper a -tile must therefore have β at that junction, so it is normal. This forces $D_{i-1,t}$.

Thus every needed local c -boundary continuation is forced by the actual rectangle and (B), rather than assumed globally.

A.5 Diagonal-chain obstruction

Diagonal-chain lemma. Assume $1 \leq j < v$, the actual $R(j-1, n+1)$ and $D_{j-1,n}$ are present, and every column with index $< j$ already has the following conditional rectangular completion property: whenever its actual component connected to the base reaches $Z_{i,t}$, the whole $R(i, t)$ is present. There cannot be an actual a - or c -edge from $V = Z_{j,n}$ in direction $e^{i\theta}$ toward OC , with its tile on the lower/right side.

Proof. Set

$$X_k = Z_{j-k, n+k}, \quad 0 \leq k \leq j.$$

The supplied $D_{j-1,n}$ gives the first actual b -edge X_0X_1 on the upper/left side. The hypothesized a - or c -edge starts at X_0 on its other side.

Suppose the actual upper/left b -chain $X_0 \cdots X_k$ has been supplied. Along each of its edges, the tiling on the opposite side supplies a collinear chain. If an opposite edge ends inside one of the existing b -edges, that b -tile has a straight sector there and forces the next actual collinear edge. Hence the opposite chain starts at X_0 and covers the entire segment X_0X_k .

If X_k were the endpoint of an opposite edge too, both sides from X_0 to X_k would be complete whole-edge chains of length kb . The opposite chain contains its initial a - or c -edge; because

$k \leq j < v$, this contradicts S_b . Therefore every intermediate X_k lies strictly inside one actual opposite edge.

At $k = 1$, put $i = j - 1$ and $t = n + 1$. The rectangle and additional tile required by the horizontal-cap lemma are exactly the supplied $R(j - 1, n + 1)$ and $D_{j-1,n}$. The lemma forces $D_{i-1,t}$, whose b -edge is $X_k X_{k+1}$.

For $i - 1 > 0$, this newly forced tile also supplies a real c -edge in column $i - 1$ from height t to $t + 1$. Its lower endpoint is already connected to the base through $R(i, t)$. The completion property of this strictly smaller column therefore supplies $R(i - 1, t + 1)$. The new $D_{i-1,t}$ is precisely the additional tile needed at the next crossing. Thus the cap can be applied successively. If any required tile leaves the target, this already contradicts the assumed tiling.

When $k = j$, the endpoint $X_j = Z_{0,n+j}$ lies on the supporting line OC . If it were beyond the side segment, an already supplied tile would have left the target. Otherwise the actual b -chain reaches the external side there. Its opposite chain cannot overrun outside the convex target, so both chains are complete from X_0 to X_j . Their length jb contradicts S_b once more. The case $j = 1$ uses this final step immediately, without an intermediate cap. $\square$

This proof permits intermediate endpoints and changes of the actual opposite tile. It requires neither $a > b$, nor a short-distance estimate, nor artificial continuation of the original a - or c -edge.

A.6 Strong column induction

Suppose $K_m = \text{conv}(0, mp, mq)$, with its standard subdivision, is present and $m \leq u < v$. By (L), its front from mp to mq is an internal chord. Its complete opposite edge chain is exactly m b -edges by S_b . At mp the old patch contributes γ , so the first opposite tile must have α there; another γ would exceed the boundary angle π . Reversing the orientation of a later opposite tile would put two γ sectors in the same half-plane at a shared endpoint. Hence every opposite tile is the normal U tile. This complete opposite row, together with K_m , supplies $R(m, 1)$.

For every actual column L_m from mp in direction q , establish:

1. its whole left side consists of normal U tiles with complete c -edges;
2. if its connected prefix reaches height t , $R(m, t)$ is present;
3. it has no right a -edge.

Column one has already been proved. Assume every smaller column is established, and maintain an actual $R(m, n)$. If L_m ends at $Z_{m,n}$, its completed normal left chain stops. Otherwise its actual upward q -ray exists.

At nq , the boundary α and γ sectors force a normal β tile $D_{0,n}$, using (B). Sweep across the top row. At each $Z_{i,n}$, $1 \leq i < m$, the old rectangle and preceding new D leave the fan θ . In the following two cases the fan is read counterclockwise from the positive horizontal ray to the θ ray.

- The order α, β gives a real long a - or c -edge in the θ direction, on its lower/right side. The diagonal-chain obstruction applies with $j = i$; the preceding sweep has supplied its required rectangle and D tile. Thus this order is impossible.
- In the order β, α , an actual upward q -ray is present and connected to column i through the old rectangle. The β tile cannot use a on that ray, by the no-right- a property of this earlier column. Hence it is normal $D_{i,n}$. The α tile cannot use b on the ray, by that earlier column's complete c -only left side. Hence it is normal $U_{i-1,n}$.

At the outer point $Z_{m,n}$, its actual continuation leaves an α fan on the left. If the α tile uses

b on the q -ray, its c -edge is a real forbidden diagonal; the just completed sweep supplies the hypotheses of the diagonal-chain obstruction with $j = m$. Thus it uses c , and is $U_{m-1,n}$. This gives $R(m, n + 1)$.

Every real continuation therefore adds a whole normal c -edge. Since the target is bounded, continuation is finite and the full component has some integer height n and the complete actual rectangle $R(m, n)$. Its boundary endpoint nq again forces one additional boundary c -edge by (B), so $n < u$. The right chain of the component has length nc ; S_c makes it c -only. In particular it contains no right a -edge. This completes the induction using only earlier columns.

A.7 Forced fronts and contradiction

Start with $K_1 = D_{0,0}$. Across its b -chord, the opposite tile is forced. The β gap at q has its boundary edge on OC and must use c there by (B). The β gap at p cannot switch to a right a -edge on L_1 . Hence K_2 is forced whenever $u \geq 2$.

Suppose K_j is present for $j \leq u < v$. By (L), its two front endpoints are strictly inside the external sides and its inner side is an actual internal chord. The complete opposite b -row is forced by S_b . Each remaining β notch admits the standard order of sides or an interchange of a, c . The notch on OC must be standard because that side has no a -edge. Any other switched notch gives a right a -edge on the actual column connected to the base through K_j and its real opposite b -row. Its column index is at most j , and all such columns have just been established from their contained K_m . Therefore no switch is possible and K_{j+1} is forced.

Thus K_{u+1} is forced. It supplies $u + 1$ complete c -edges on OC , contradicting its exact count u from (B). This concludes the proof.

Scope and relation to parametrization

The two obstructions apply to the full primitive scale-one families stated above. If a tiling in one of these families is parametrized with an integer scale t and tile count $t^2(3v^2 - u^2)$ or $t^2(2v^2 - u^2)$, then a squarefree tile count forces $t = 1$ and is excluded by the corresponding obstruction. This is a consequence within these two parametrized branches only; it does not classify all squarefree tile counts or all composite tile counts in Erdős problem 634.

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